

Proof of the VC-Generalization Bound

E_{out} is the error in the input space X (Risk)

E_{in} is the error on the data set D (Empirical Risk)

E'_{in} is the error on the independent data set D' .

Let D and D' have sample size N .

$\mathcal{H}$ is the hypothesis set where $h \in \mathcal{H} \Rightarrow h: X \rightarrow \{-1, 1\}$.

Dichotomies generated by $\mathcal{H}$:

$$\mathcal{H}(x_1, \dots, x_N) = \left\{ (h(x_1), \dots, h(x_N)) : h \in \mathcal{H} \right\}$$

$$m_{\mathcal{H}}(N) = \max_{x_1, \dots, x_N \in X} |\mathcal{H}(x_1, \dots, x_N)|$$

where $|A|$ is the cardinality of the set A .

We will prove:

Theorem: (Vapnik, Chervonenkis - 1971)

$$\mathbb{P} \left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \varepsilon \right] \leq 4 m_{\mathcal{H}}(2N) e^{-\frac{1}{8} \varepsilon^2 N}.$$

Lemma:

$$(1 - 2e^{-\frac{1}{2}\epsilon^2 N}) \mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \epsilon\right] \\ \leq \mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \epsilon/2\right].$$

where the probability on the right hand side is over $\mathcal{D}$ and $\mathcal{D}'$ jointly.

Proof:

WLOG assume $\mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \epsilon\right] > 0$, otherwise there is nothing to prove.

$$\mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \epsilon/2\right]$$

$$* \geq \mathbb{P}\left[\underbrace{\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \epsilon/2}_{B_1} \text{ and } \underbrace{\sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \epsilon}_{B_2}\right]$$

$$** = \mathbb{P}[B_2] \cdot \mathbb{P}[B_1 | B_2]$$

* $\mathbb{P}[B_1] \geq \mathbb{P}[B_1 \text{ and } B_2]$ for any events B_1, B_2

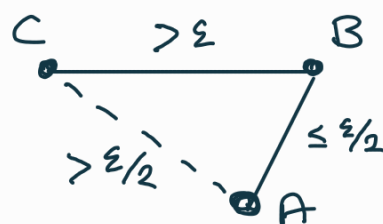
** $\mathbb{P}[B_1 \text{ and } B_2] = \mathbb{P}[B_2] \cdot \mathbb{P}[B_1 | B_2]$ for any B_1, B_2 .

By our assumption $\exists h^* \in \mathcal{H}$ s.t. $|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon$
 where h^* depends on D but is independent of D' .

Then, we have

$$\begin{aligned}
 & \mathbb{P} \left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid \sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \varepsilon \right] \\
 & \geq \mathbb{P} \left[|E_{in}(h^*) - E'_{in}(h^*)| > \varepsilon/2 \mid \sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \varepsilon \right] \\
 * & \geq \mathbb{P} \left[|E'_{in}(h^*) - E_{out}(h^*)| \leq \varepsilon/2 \mid \sup_{h \in \mathcal{H}} |E_{in}(h) - E_{out}(h)| > \varepsilon \right] \\
 ** & \geq 1 - 2e^{-\frac{1}{2}\varepsilon^2 N} \quad \blacksquare
 \end{aligned}$$

* follows from the fact that



$$\underbrace{|E'_{in}(h^*)|}_A - \underbrace{E_{out}(h^*)}_B \leq \varepsilon/2 \text{ and } \underbrace{|E_{in}(h^*)|}_C - \underbrace{E_{out}(h^*)}_B > \varepsilon \text{ implies }$$

$$\underbrace{|E_{in}(h^*)|}_C - \underbrace{|E'_{in}(h^*)|}_A > \varepsilon/2.$$

** follows from the Hoeffding's inequality:

$$\mathbb{P} \left[|E'_{in}(h^*) - E_{out}(h^*)| > \varepsilon \right] \leq 2e^{-2\varepsilon^2 N}$$

since h^* is independent of D' (i.e. h^* is fixed w.r.t. D').

Note: We can assume that $e^{-\frac{1}{2}\varepsilon^2 N} < \frac{1}{4}$ otherwise

$$e^{-\frac{1}{8}\varepsilon^2 N} = \left(e^{-\frac{1}{2}\varepsilon^2 N} \right)^{1/4} \geq \left(\frac{1}{4} \right)^{1/4} = \frac{1}{\sqrt{2}}.$$

Then the theorem is trivially true. Now, by assumption

$$\left(1 - 2 e^{-\frac{1}{2}\varepsilon^2 N} \right) > \frac{1}{2} \text{ and then the lemma implies}$$

that

$$\mathbb{P} \left[\sup_{h \in \mathcal{H}} |E_{\text{in}}(h) - E_{\text{out}}(h)| > \varepsilon \right] \leq 2 \mathbb{P} \left[\sup_{h \in \mathcal{H}} |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \varepsilon/2 \right].$$

Lemma: $\mathbb{P} \left[\sup_{h \in \mathcal{H}} |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \varepsilon/2 \right]$

$$\leq m_{\mathcal{H}}(2N) \sup_S \sup_{h \in \mathcal{H}} \mathbb{P} \left[|E_{\text{in}}(h) - E'_{\text{in}}(h)| > \varepsilon/2 \mid S \right]$$

where the probability on the LHS is over D and D' jointly and the probability on the RHS is over random partitions of sample S of size $2N$ into two sets D and D' of size N .

Proof: Sample S with size $2N$, then sample again $D \subset S$ with size N . let $D' = S \setminus D$. Then by the law of total probability (by $\sum$ denoting sum or integral as the case may be),

$$\begin{aligned} \mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2\right] \\ = \sum_S \mathbb{P}[S] \cdot \mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right] \end{aligned}$$

$$\left(\text{since } \sum_S \mathbb{P}[S] = 1\right) \rightarrow \leq \sup_S \mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right].$$

let $\mathcal{H}(S)$ be the dichotomies that $\mathcal{H}$ can implement on the points in S . Then $M := |\mathcal{H}(S)| \leq m_{\mathcal{H}}(2N)$.

Let $\mathcal{H}(S)$ be realized by $h_1, \dots, h_M$. Thus,

$$\mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right] = \mathbb{P}\left[\sup_{h \in \{h_1, \dots, h_M\}} |E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right]$$

$$\text{union bound} \rightarrow \leq \sum_{n=1}^M \mathbb{P}\left[|E_{in}(h_n) - E'_{in}(h_n)| > \varepsilon/2 \mid S\right]$$

$$\leq M \cdot \sup_{h \in \mathcal{H}} \mathbb{P}\left[|E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right].$$

Lemma 1: (Hoeffding, 1963)

Let $A = \{a_1, \dots, a_{2N}\}$ be a set of values with $a_n \in [0, 1]$, and let $\mu = \frac{1}{2N} \sum_{n=1}^{2N} a_n$ be their mean. Let $D = \{z_1, \dots, z_N\}$ be a sample of size N , sampled from A uniformly without replacement. Then,

$$\mathbb{P}\left[\left|\frac{1}{N} \sum_{n=1}^N z_n - \mu\right| > \varepsilon\right] \leq 2 e^{-2\varepsilon^2 N}.$$

Lemma 2: For any h and S ,

$$\mathbb{P}\left[|E_{in}(h) - E'_{in}(h)| > \varepsilon/2 \mid S\right] \leq 2 e^{-\frac{1}{8} \varepsilon^2 N},$$

where the probability is over random partitions of S into two sets D and D' .

Proof: We apply lemma 1 as follows: For the $2N$ samples in S , let $a_n = 1$ if $h(x_n) \neq y_n$ and $a_n = 0$ otherwise. Then $\{a_n\}$ are the errors made by h on S . Then

$$E_{in}(h) = \frac{1}{N} \sum_{a_n \in D} a_n \quad \text{and} \quad E'_{in}(h) = \frac{1}{N} \sum_{a'_n \in D'} a'_n$$

$S = D \cup D'$ and $D \cap D' = \emptyset$. Therefore

$$\mu = \frac{1}{2N} \sum_{n=1}^{2N} a_n = \frac{E_{in}(h) + E'_{in}(h)}{2}.$$

Then, it follows that

$$|E_{in}(h) - \mu| > t \iff |E_{in}(h) - E'_{in}(h)| > 2t$$

$$\begin{aligned} \text{since } |E_{in}(h) - E'_{in}(h)| &= |E_{in}(h) - 2\mu + E_{in}(h)| \\ &= 2|E_{in}(h) - \mu|. \end{aligned}$$

So by Lemma 2, we have

$$\mathbb{P}[|E_{in}(h) - E'_{in}(h)| > 2t] = \mathbb{P}[|E_{in}(h) - \mu| > t] \leq 2e^{-2t^2N}.$$

Substituting $t = \varepsilon/4$ gives the result. 

Note: We showed that

$$\mathbb{P}\left[\sup_{h \in \mathcal{H}} |E_{\text{in}}(h) - E_{\text{out}}(h)| > \varepsilon\right] \leq 4 m_{\mathcal{H}}(2N) e^{-\frac{1}{8} \varepsilon^2 N}.$$

In other words, if we let $\delta := 4 m_{\mathcal{H}}(2N) e^{-\frac{1}{8} \varepsilon^2 N}$, then with probability $1 - \delta$, we have that

$$\sup_{h \in \mathcal{H}} |E_{\text{out}}(h) - E_{\text{in}}(h)| < \varepsilon.$$

Therefore, for any $g \in \mathcal{H}$

$$E_{\text{out}}(g) - E_{\text{in}}(g) \leq \sup_{h \in \mathcal{H}} |E_{\text{out}}(h) - E_{\text{in}}(h)| < \varepsilon.$$

Expressing ε in terms of δ gives:

Theorem: For any $\delta > 0$ and any $g \in \mathcal{H}$, we have

$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \sqrt{\frac{8}{N} \ln\left(\frac{4 m_{\mathcal{H}}(2N)}{\delta}\right)}$$

with probability $1 - \delta$.